

SESSION - 06

Templates



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TEMPLATES

What is a template?

Templates are used to create pages. When an author creates a page, the nodes and properties of the template are copied to the location of the new page. When pages are created, the author must choose a template to base the page on.

When the *sling:resourceType* property of the template is copied to the new page location a connection is made between the page and its rendering component. Other properties of the template can control where the template may be used and even the order it will appear in the Create Page dialog.

The template can also contain initial content for the page in a *jcr:content* node tree.

Properties of a Template

AEM templates have the following attributes and properties (not a complete list but includes the most important properties):

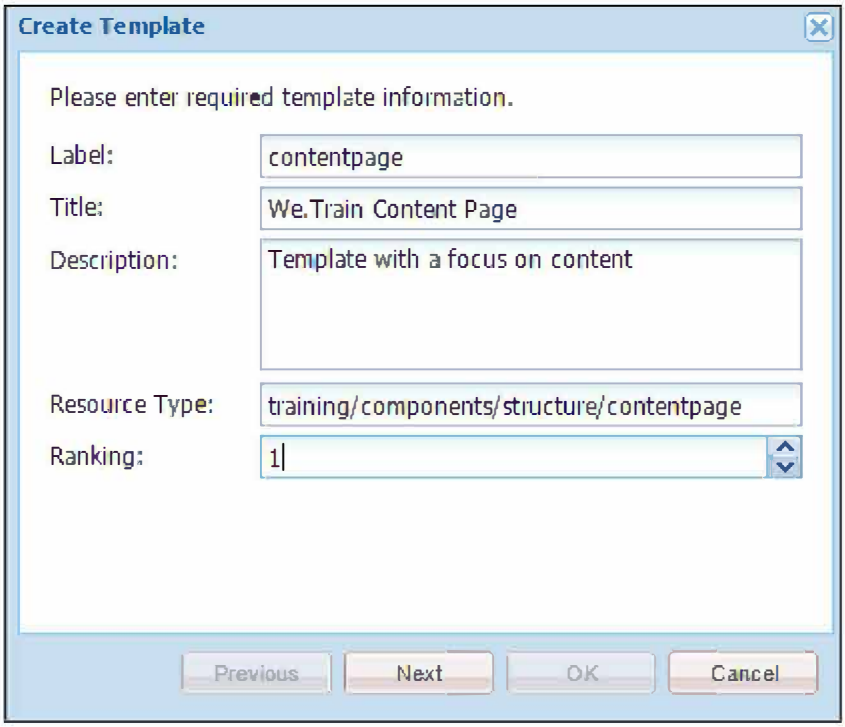
- label – becomes the node name
- sling:resourceType – defines the set of scripts that can be used to render the page
- jcr:title – the string used by the Consoles to display this template
- jcr:description – documentation/description for this template
- ranking – defines the order in which this template will appear, as compared to other available templates in the "Create" dialog
- allowedPaths – paths where this template can be used
- allowedParents – paths of templates that are allowed to be parent to this template
- allowedChildren – paths of templates that are allowed to be children to this template
- thumbnail.png – a thumbnail image that helps the author identify this template

Exercise - I

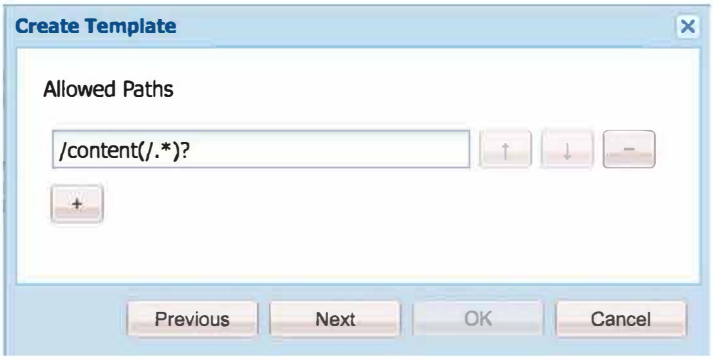
Templates play many parts within the AEM ecosystem. One important part is to be in the list of available templates when the content author creates a page.

Task – Create a Template

1. **In CRXDE Lite, navigate to /apps/training/templates.**
2. **Right click /apps/training/templates and select Create…> Create Template.**
3. **Enter the following values:**
   - **Label**: contentpage
   - **Title**: We.Train Content Page
   - **Description**: Template with a focus on content
   - **Resource Type**: training/components/structure/contentpage
   - Ranking: 1



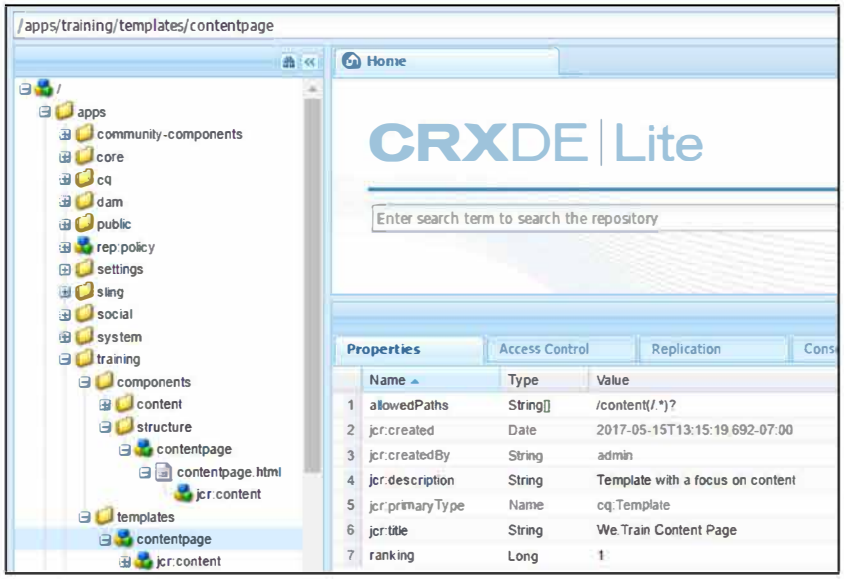
4. **Click Next.**
5. **Click the + and enter /content(/.*)? for Allowed Paths.**



6. **Click Next.**
   NOTE: Do not click on the "+" for Allowed Parents.

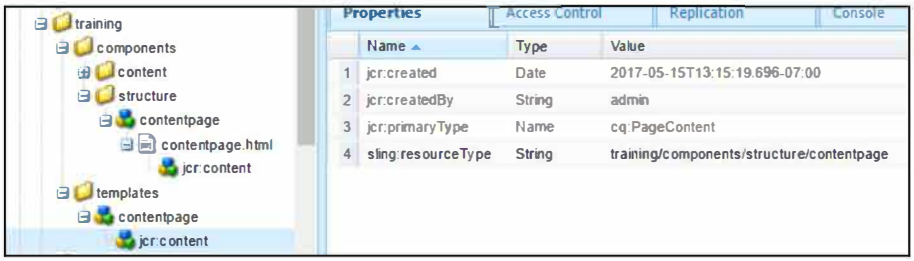
7. **Click Next.**
   NOTE: Do not click on the "+" for Allowed Children.
8. **Click OK.**
9. **Save all changes.**

Your project node structure should look like this:



10. Take a look at the properties that were created on the /apps/training/templates/contentpage node:
    - allowedPaths property specifies where the template can be used to create pages and came from the Allowed Paths pane in the Create Template wizard.
    - jcr:description came from the Description entry in the Create Template wizard.
    - jcr:title came from the Title entry in the Create Template wizard
    - ranking property specifies where in the list of templates this template will appear and came from the Ranking entry in the Create Template wizard. Ours will be first, as we have assigned a ranking of "1".

Notice that the value from the Resource Type entry in the Create Template wizard does not appear. This value becomes the sling:resourceType property, which does not appear at this level. We will find the sling:resourceType property on the jcr:content child node of the contentpage template node.

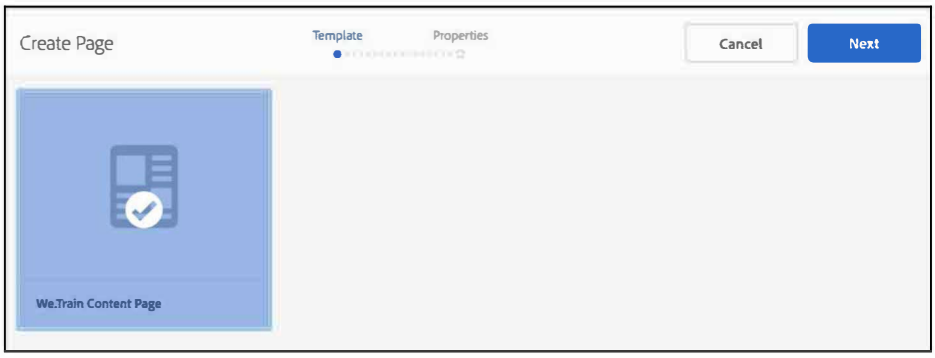


Task – Test the contentpage template

1. **Navigate to the Sites Console by entering http://localhost:4502/sites.html or by typing "/", entering "sites" into the OmniSearch toolbar, and choosing Go to Sites.**
2. **Click Create and choose Page.**



3. **Select the We.Train Content Page template and click Next.**

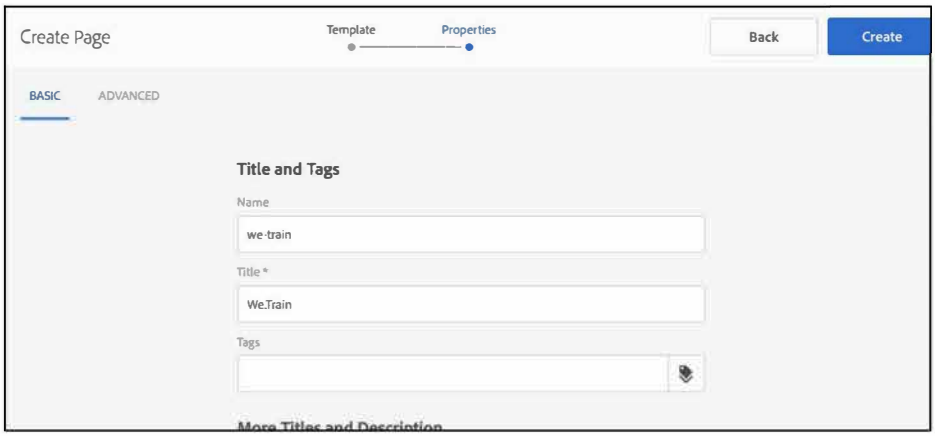


Troubleshooting tip: If your template does not appear in the list, check the allowedPaths property on the /apps/training/templates/contentpage template. The value should be: "**/content(/.*)?**". Any other value will prohibit the template from appearing in the list of available templates.

4. **Enter the following properties:**

Name: we-train

Title: We.Train



5. **Click Create.**
6. **Click Open (This opens the page in Edit mode). It should appear like this:**



Troubleshooting tip: If your page is blank, check the **sling:resourceType** property on the /apps/training/templates/contentpage/jcr:content node. It should exactly match the path to the contentpage page-rendering component. For example, the relative path might be training/components/structure/contentpage, or an absolute path might be /apps/training/components/structure/contentpage.

If the sling:resourceType path is incorrect, Apache Sling will not be able to find a rendering script and that is why you are seeing a blank page. Correcting the sling:resourceType property on the template will correct the problem for all new pages. Now you have two choices: 1) you can delete the page you created and recreate it or 2) correct the sling:resourceType property on the jcr:content child of the page you created.

Pro Tip: When a blank page or blank component occurs, look in the error log, located at <AEM instance root>/crx-quickstart/logs/error.log. You will find an entry for the request that states "... could not find renderer for "<path to the request>. This error message tells you that Sling could not find a rendering script for that request. Carefully examine the path specified in the error message and then look in the repository to fix the problem.

Task – Restrict Template Use

You can restrict where a template can be used in two ways. We have already seen that the allowedPaths property on the /apps/training/templates/contentpage template will define the paths where this template can be used to create pages. This restricts the template from the template's point of view.

1. **Navigate to the AEM Sites Console. Click Create Page.**
2. **Notice that the We.Train template appears.**
3. **Do the same with Sites>Screens. Notice the We.Train template appears.**

NOTE: We probably want the We.Train template to appear as an available template only for the We.Train site.

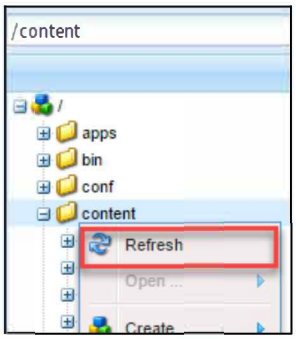
4. **Using CRXDE Lite, navigate to /apps/training/templates/contentpage.**
5. **Modify the value of the allowedPaths property (by double-clicking on this property) to be:**

   /content/we-train(/.*)?

6. **Repeat steps 1-3. Notice that this time the We.Train template no longer appears as a choice for Screens.**
7. **Navigate to We.Train.**
8. **Click Create Page. Notice that the We.Train Content Page template is available.**

Another way to restrict templates is to specify which templates may be used within a specified path. We can accomplish this using the cq:allowedTemplates property on a page.

9. **Refresh CRXDE Lite by clicking on the /content folder and choosing Refresh:**



10. **Using CRXDE Lite, navigate to /content/we-train/jcr:content.**
11. **Add the following property:**

    - Name: cq:allowedTemplates
    - Type: String[]
    - Value: /apps/training/templates/.*

    Tip: To enter a String Array (String[]), choose String and click the Multi button. Beware! The Multi button stays clicked until you unclick it.



12. **Save your changes.**
13. **Test out the restriction by going to the AEM Sites Console. Using the same logic as we used previously in this task, make sure that the We.Train template is still available within the We.Train site.**

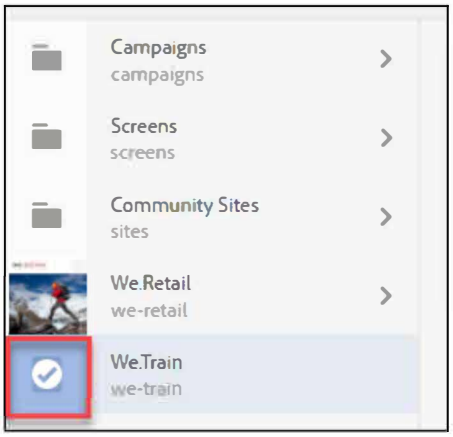
These two mechanisms allow us to control where templates are made available for use. The allowedPaths property allows the developer to decide where to use templates. The cq:allowedTemplates property allows the website owner to decide which templates to use within the site.

Task – Add Content Structure to the Template

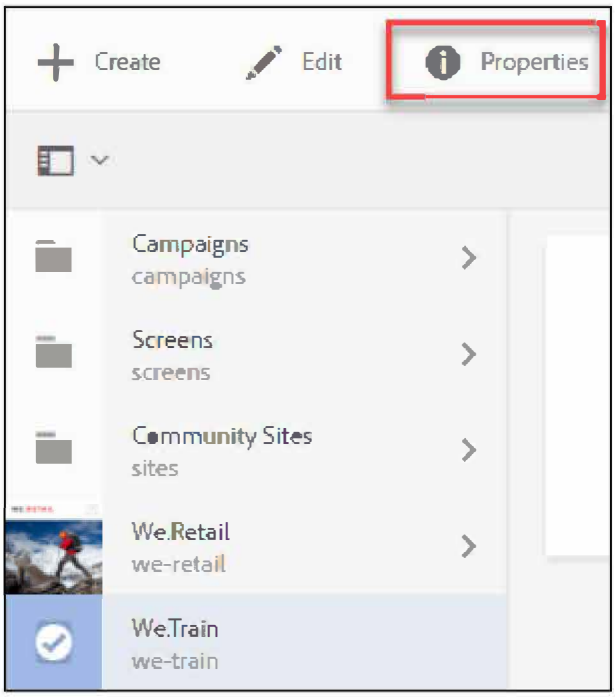
As mentioned earlier, the template plays many roles. One role that we experienced earlier is to appear in the list of available templates when creating a page. Another role that the template plays is to "front-load" content onto a page created from that template.

So, let's say we wanted every page created by the contentpage template in the We-train site to have the same thumbnail image. Here's how to achieve this:

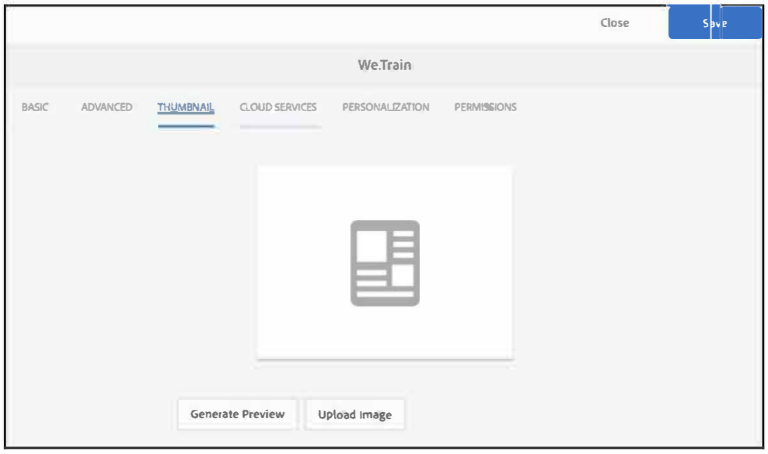
1. **Using the Sites Console, navigate to We.Train.**
2. **Select We.Train:**



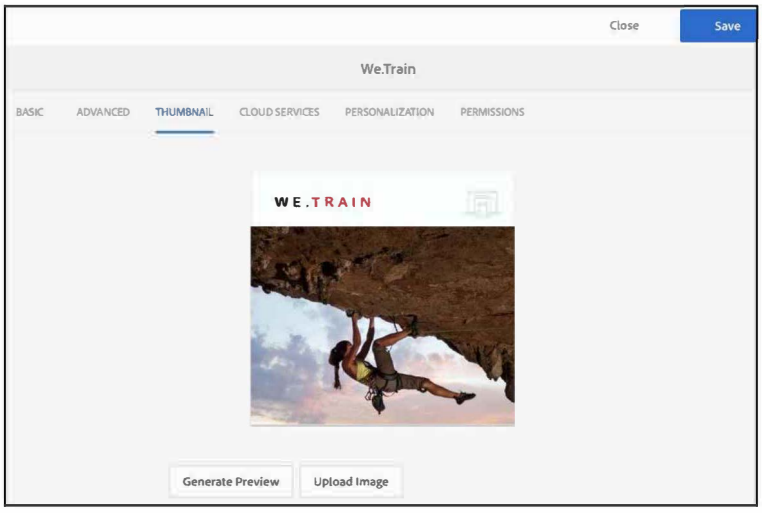
3. **Click Properties.**



4. **Click the Thumbnail tab.**

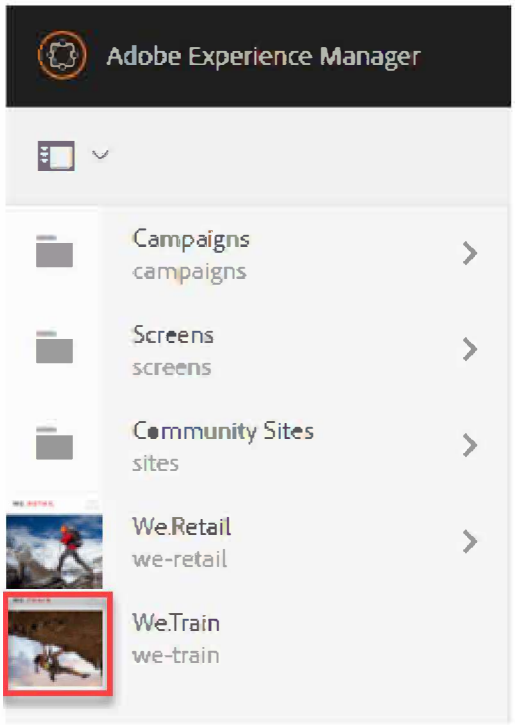


5. **Click on Upload Image and upload We-train.png from the exercise files provided to you for this exercise into the thumbnail.**



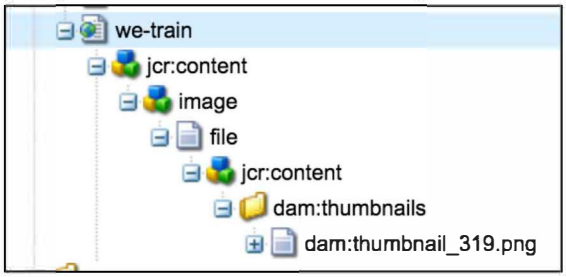
6. **Click Save & Close in the upper right.**

7. **Notice that your Site now has a thumbnail image associated with it:**



8. **Now we have created the thumbnail structure under the /content/we-train page that we want to have on every page. Let's modify the definition of the /apps/training/templates/contentpage template so that every page created from that template has this thumbnail.**
9. **Using CRXDE Lite, navigate to /content/we-train/jcr:content.**

   Notice the newly defined structure under the jcr:content node. **NOTE:** You may need to refresh CRXDE Lite.

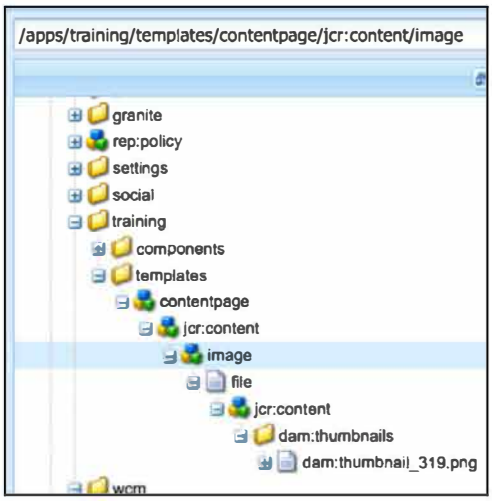


10. **Right-click on the image node and select Copy.**

NOTE: Do not copy the "dam:thumbnail_319.png" node. Instead, ensure you are copying the image node directly under jcr:content.

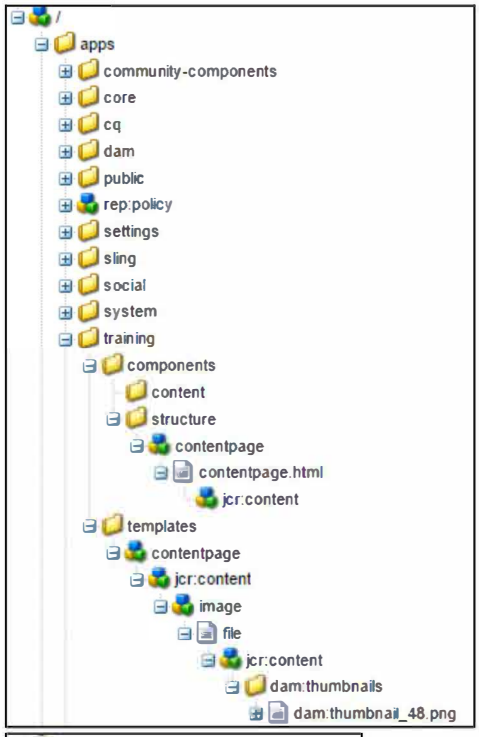
11. **Navigate to /apps/training/templates/contentpage/jcr:content.**

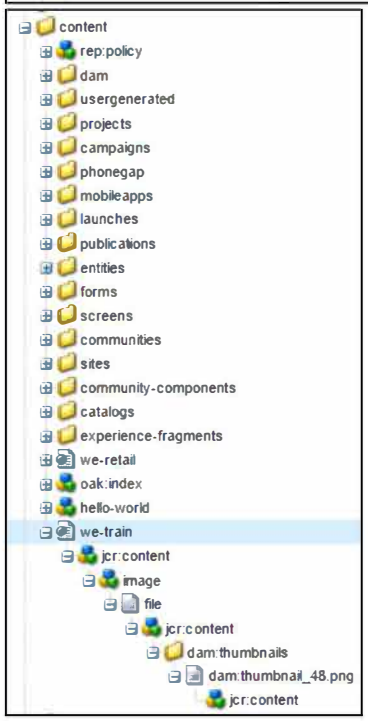
12. **Right-click the jcr:content node and select Paste. Your node structure should now look like this (NOTE: The "319" number at the end of the *.png filename may be different in your instance):**



NOTE: When a page is created from a template, AEM writes the cq:Page parent and then copies the template's jcr:content child as the child to the cq:Page node. Now, every page we create from the contentpage template will have the thumbnail.

NOTE: Your /apps and /content node structures (respectively) should look like this in CRXDE Lite:





Creating the Website Structure

At a very basic level, creating a page structure is how a website structure is set up. A page is where content authors create and edit content that most likely will be published and viewed by site visitors.

A page is many things:

- Website content container
- Instance of a template
- cq:Page JCR Node type (has a mandatory jcr:content child node)

When creating a page, the content that authors enter in the dialog box becomes the nodes and associated properties for that page. The Page Creation wizard allows you to enter or select the following information, which is necessary to create the complete page structure.

- Name: cq:Page node name
- Title: jcr:title property
- Template: cq:template property

The template's *sling:resourceType* property is added as the page's *sling:resourceType* property, so the page knows where its rendering script is. When working with pages, you can use the following WCM APIs:

- com.day.cq.wcm.api.Page
- com.day.cq.wcm.api.PageManager
- com.day.cq.wcm.api.PageFilter

Exercise – II

Task – Create the pages for the site

Now it is time to create our website structure. It won't be a very interesting website yet, as all the pages will be identical. But we will continue to build out our page-rendering component and create other components to fill out the page content.

1. **Using the Sites Console, navigate to We.Train.**
2. **Click Create Page.**
3. **Choose the We.Train template and click Next.**
4. **Fill in the Name and Title:**

- Name: en
- Title: English

(This page will be the English language root.)

5. **Click Create, and then click Done.**
6. **At the same level, create the French language root page:**
    - Name: fr
    - Title: Francais

Notice the thumbnail image appearing under the English and French language roots:



This came from the content that we placed under the template in the last exercise.

7. **Under the English root, create the following website page structure.**

NOTE: Use a naming best practice by naming all your pages in *lowercase* and use hyphens for two or more words:

en (Title: English) >

- about (Title: About)
- communities (Title: Communities)
- experiences (TItle: Experiences)
- products (TItle: Products)>
    - biking (Title: Biking)
    - hiking (Title: Hiking)
    - running (Title: Running)
    - skiing (Title: Skiing)

NOTE: Your page structure should look like this when complete:

